13/03/2026 · Linear Algebra — [CP]

Q: Price of Linear Algebra Book = x
Price of Differential Equation Book = y
Price of Educational Psychology Book = z

- Asma Buys: $x + y = \$178$
- Aiman Buys: $2x + y + z = \$319$
- Sara Buys: $z + x = \$147$

Augmented Matrix:
$$\left[\begin{array}{ccc:c} 1 & 1 & 0 & 178 \\ 2 & 1 & 1 & 319 \\ 1 & 0 & 1 & 147 \end{array}\right]$$

$$\Rightarrow R_2 - 2R_3: \left[\begin{array}{ccc:c} 1 & 1 & 0 & 178 \\ 0 & 1 & -1 & 25 \\ 1 & 0 & 1 & 147 \end{array}\right] \Rightarrow R_3 - R_1: \left[\begin{array}{ccc:c} 1 & 1 & 0 & 178 \\ 0 & 1 & -1 & 25 \\ 0 & -1 & 1 & -31 \end{array}\right]$$

$$\Rightarrow R_3 + R_1: \left[\begin{array}{ccc:c} 1 & 1 & 0 & 178 \\ 0 & 1 & -1 & 25 \\ 0 & 0 & 1 & 147 \end{array}\right] \Rightarrow R_2 + R_3: \left[\begin{array}{ccc:c} 1 & 1 & 0 & 178 \\ 0 & 1 & 0 & 172 \\ 0 & 0 & 1 & 147 \end{array}\right]$$

$$\Rightarrow R_1 - R_2: \left[\begin{array}{ccc:c} 1 & 0 & 0 & 6 \\ 0 & 1 & 0 & 172 \\ 0 & 0 & 1 & 147 \end{array}\right]$$

$x = \$6$
$y = \$172$
$z = \$147$

$\Rightarrow$ Asma Buys: $x + y = \$178 \Rightarrow \$6 + \$172 = \$178$
$\Rightarrow$
$\Rightarrow$